APPLE SQL

Interview Questions

for Data Analyst



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1. Trade-In Revenue by Store

Problem Statement: Apple’s trade-in program allows customers to return their old iPhones for cash. Each store needs to know how much revenue they made from these trade-ins. We want to see the total payouts for each store, sorted from highest to lowest.

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trade_in_transactions

Column Name	Type
transaction_id	integer
model_id	integer
store_id	integer
transaction_date	date

trade_in_payouts

Column Name	Type
model_id	integer
model_name	string
payout_amount	integer

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How to Solve:

- Use the trade_in_transactions table for transaction records and the trade_in_payouts table for payout amounts.
- Connect the two tables using the model_id to ensure you get the correct payout for each transaction.
- Use GROUP BY to organize the results based on each store’s ID.

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- Use SUM() to calculate the total payout amount for each store.

- Use ORDER BY to arrange the stores in descending order of their total payouts.

- Choose the columns to display: the store ID and the calculated total payout.

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SQL

```sql
SELECT
  t.store_id,
  SUM(p.payout_amount) AS total_revenue
FROM trade_in_transactions AS t
INNER JOIN trade_in_payouts AS p
  ON t.model_id = p.model_id
GROUP BY t.store_id
ORDER BY total_revenue DESC;
```

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2. AirPods Purchase Patterns After iPhones

Apple Statement: The Apple retention team wants to understand buying patterns. We need to find the percentage of customers who purchased AirPods immediately after buying iPhones. This includes customers who bought both at the same time. We must round the final percentage to the nearest whole number.

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transactions

Column Name	Type
transaction_id	integer
customer_id	integer
product_name	varchar
transaction_timestamp	datetime

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How to Solve:

- Use the transactions table to analyze customer purchases.

- Apply LAG() to check if AirPods were bought directly after iPhones for each customer.

- Select only those customers who bought AirPods following iPhones, without any other purchases in between.

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- Calculate the number of unique customers who fit the criteria.

- Find the total number of distinct customers who made any purchase.

- Use the formula (number of AirPods buyers / total customers) * 100.

- Round the final percentage to ensure it’s a whole number.

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SQL

```sql
WITH lag_products AS (
  SELECT
    customer_id,
    product_name,
    LAG(product_name) OVER(PARTITION BY customer_id ORDER BY
transaction_timestamp) AS prev_prod
  FROM transactions
),
interested_users AS (
  SELECT DISTINCT customer_id AS airpod_iphone_buyers
  FROM lag_products
  WHERE LOWER(product_name) = 'airpods'
    AND LOWER(prev_prod) = 'iphone'
)

SELECT
  ROUND(
    COUNT(DISTINCT iu.airpod_iphone_buyers)::DECIMAL
    / COUNT(DISTINCT transactions.customer_id)::DECIMAL
    * 100, 0
  ) AS follow_up_percentage
FROM transactions
LEFT JOIN interested_users AS iu
  ON iu.airpod_iphone_buyers = transactions.customer_id;
```

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3. Product Ratings by Month

Apple Statement: We need to analyze customer reviews for Apple products. The goal is to calculate the monthly average rating for each product based on user reviews. Each review includes a star rating from 1 to 5.

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The reviews table has the following schema:

- review_id: An integer that uniquely identifies each review
- user_id: An integer that identifies the user who submitted the review
- submit_date: A datetime value that represents when the review was submitted
- product_id: An integer that identifies the product being reviewed
- stars: An integer that indicates the number of stars the product received in the review (from 1 to 5)

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How to Solve:

- Use the reviews table, which contains all necessary data.
- Use the EXTRACT() function to pull the month from the submit_date.
- Organize results by month and product ID to calculate averages.
- Use the AVG() function to determine the average star rating for each product each month.
- Sort Results: Order the results first by month and then by product ID for clarity.

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SQL

```sql
SELECT
  EXTRACT(MONTH FROM submit_date) AS mth,
  product_id AS product,
  AVG(stars) AS avg_stars
FROM reviews
GROUP BY mth, product
ORDER BY mth, product;
```

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4. Average Monthly Sales of Apple Products

Apple Statement: As a data analyst at Apple, understanding the sales performance of various products is crucial. We need to calculate the average quantity sold per month for each Apple product during the year 2021. This data will help in making informed business decisions and creating insightful visualizations.

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products

product_id	product_name
1	iPhone 12
2	Apple Watch
3	MacBook Pro

sales

sales_id	product_id	date_of_sale	quantity_sold
1	1	1/10/2021	100
2	1	1/15/2021	200
3	2	1/20/2021	50
4	2	2/15/2021	75
5	3	2/10/2021	20

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How to Solve:

- Use the products table for product details and the sales table for sales data.

- Perform an inner join on product_id to combine sales data with product names.

- Use a WHERE clause to focus on sales that occurred in 2021.

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- Use the MONTH() function to group sales data by month.

- Apply the AVG() function to determine the average quantity sold for each product each month.

- Group the results by month and product name for clear analysis.

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SQL

```sql
SELECT
  MONTH(s.date_of_sale) AS 'Month',
  p.product_name,
  AVG(s.quantity_sold) AS 'Average_Sold'
FROM sales s
JOIN products p ON s.product_id = p.product_id
WHERE YEAR(s.date_of_sale) = 2021
GROUP BY Month, p.product_name;
```

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